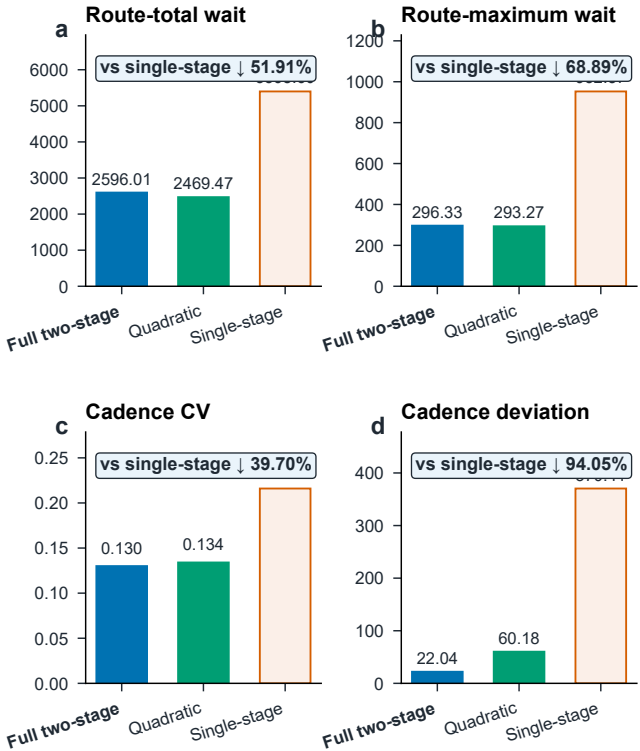Same 600-s budget: the full two-stage workflow improves all four operational metrics over single-stage